ExplainableCropAI: Multi-level Explanation Process

Input Image

Field-captured
paddy leaf RGB
640 × 640 × 3

RGB image

ConvNeXt V2-Tiny — NOVEL

Modern CNN backbone
for feature extraction

Multi-scale features (P2–P5)

Top features (P5)

Visual Explanation — NOVEL

Grad-CAM++ attribution
Second-order gradients
produce heatmaps over
disease-relevant regions

Uncertainty

Monte Carlo Dropout
30 stochastic passes
split into epistemic
and aleatoric components

Visual evidence

Prediction + uncertainty

Phi-3-mini LLM — NOVEL

Audience-adaptive natural-
language generation
fed with prediction,
uncertainty and visual
evidence

Diagnosis context

Recommendation Engine

Context-aware treatment
guidance, conditioned on
diagnosis, uncertainty
and agronomic priors

Agronomic rules

Knowledge Base

Agronomic rules,
treatments, severity
thresholds

Explanation

Treatment

Decision Support Output

Three synchronised channels:
prediction + confidence | visual explanation | NL explanation + recommendations

Deployed insights

End User / Mobile App

Farmer or
extension officer

Components

- Input / Output
- CNN Backbone
- Visual Explanation
- Uncertainty
- LLM Module
- Recommender
- Knowledge Base
- N — Novel contribution